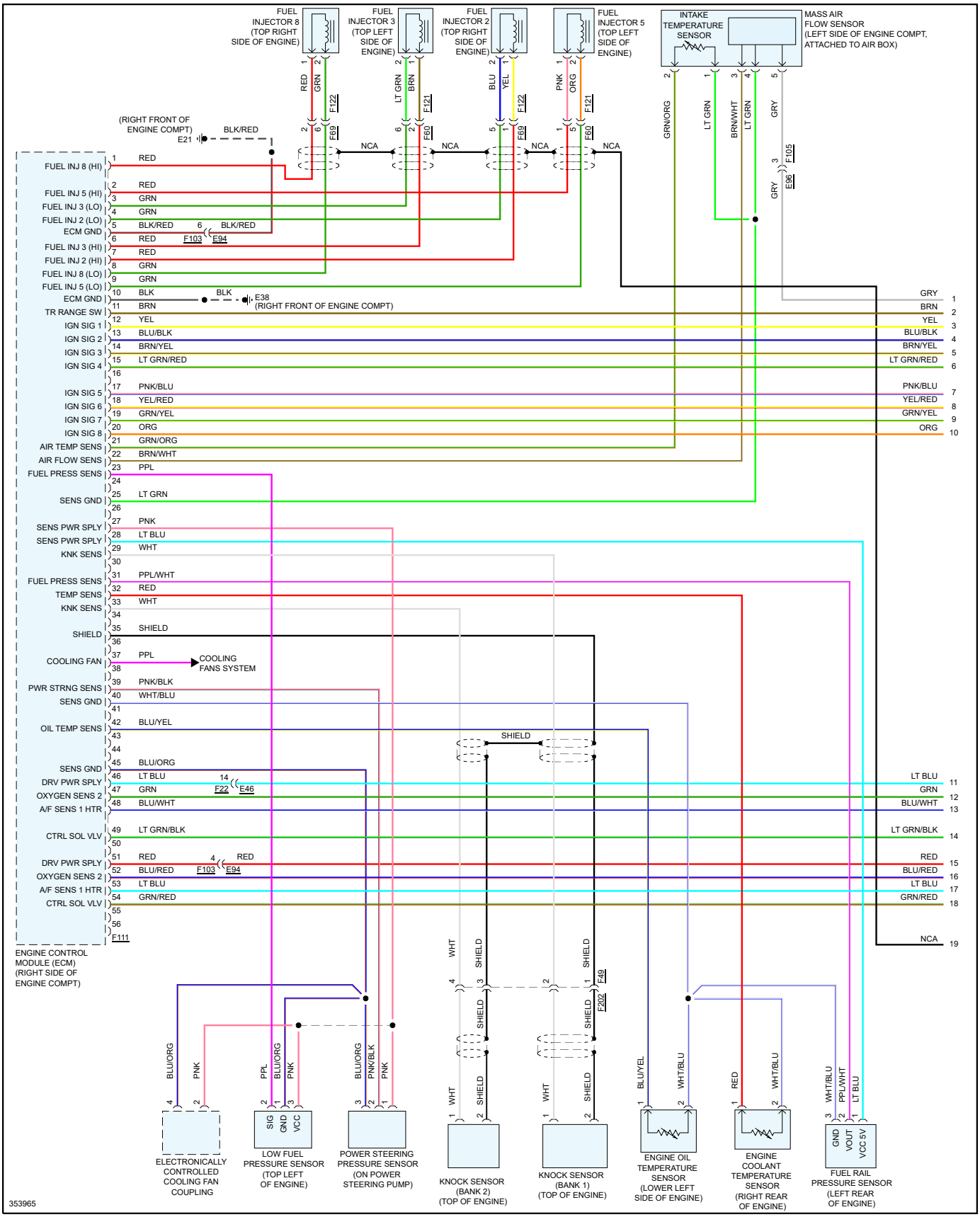
2011 Infiniti Truck QX56 AWD V8-5.6L (VK56VD)

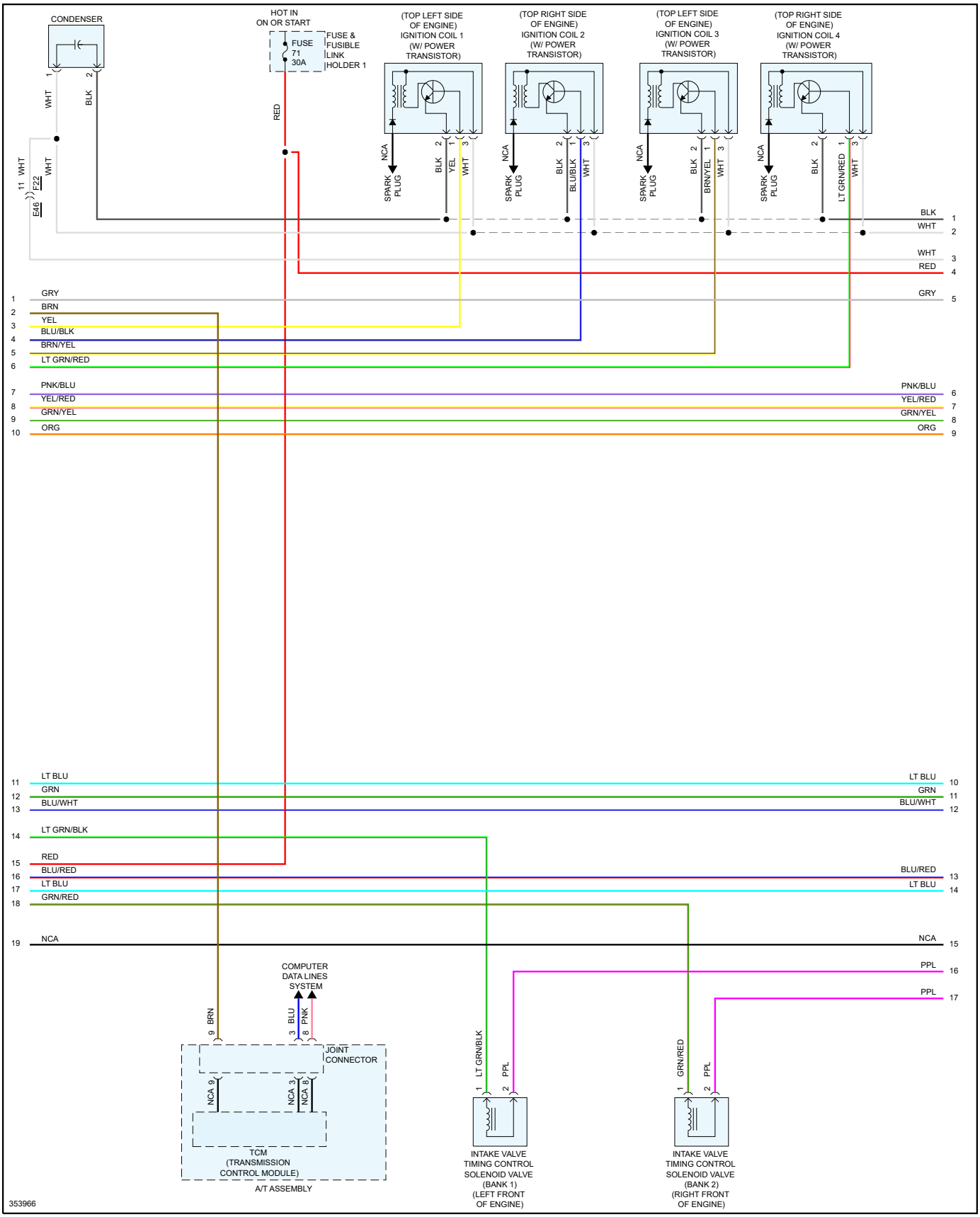
Vehicle > Powertrain Management > Diagrams > Electrical - Interactive Color (Non OE)

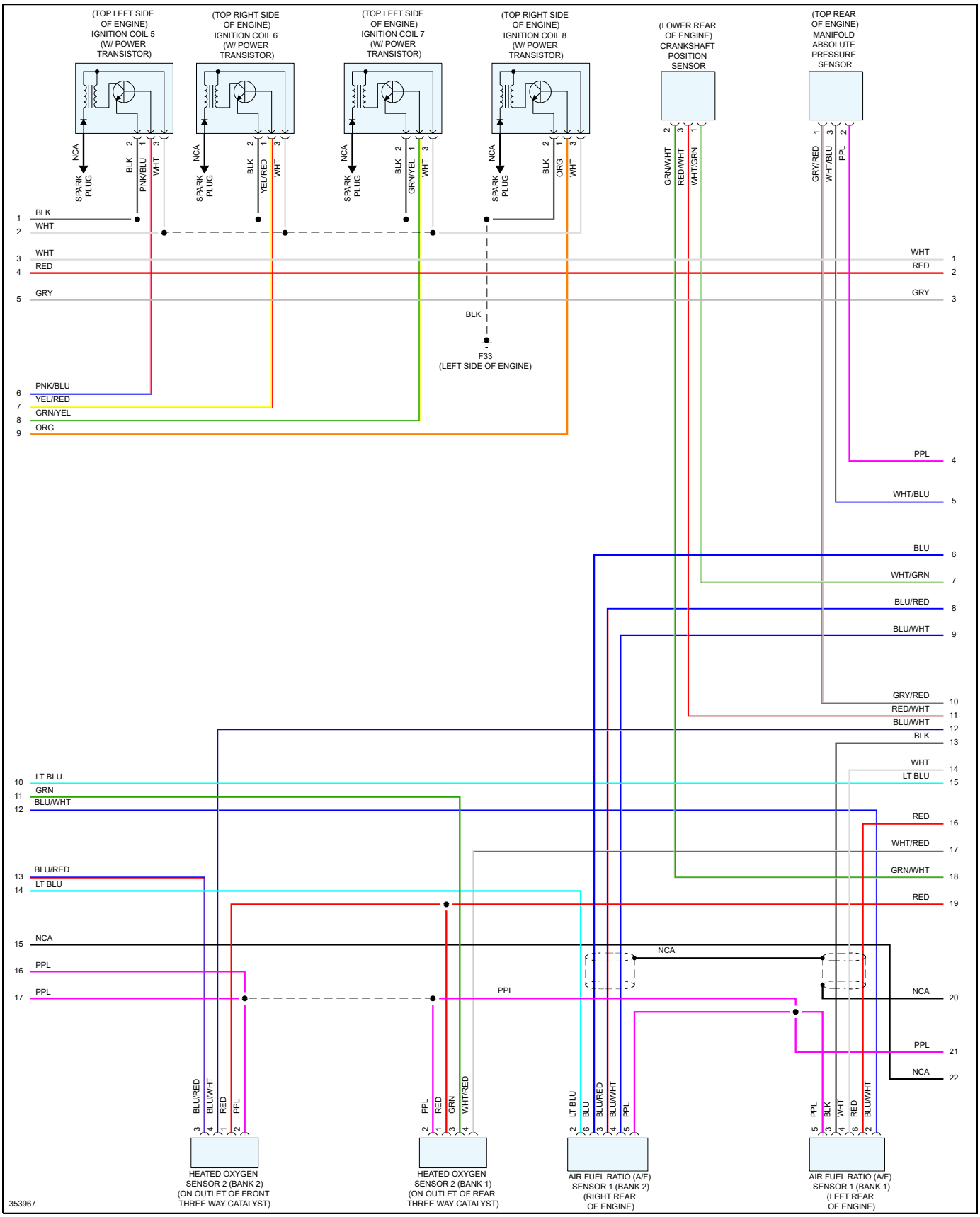
ENGINE CONTROLS

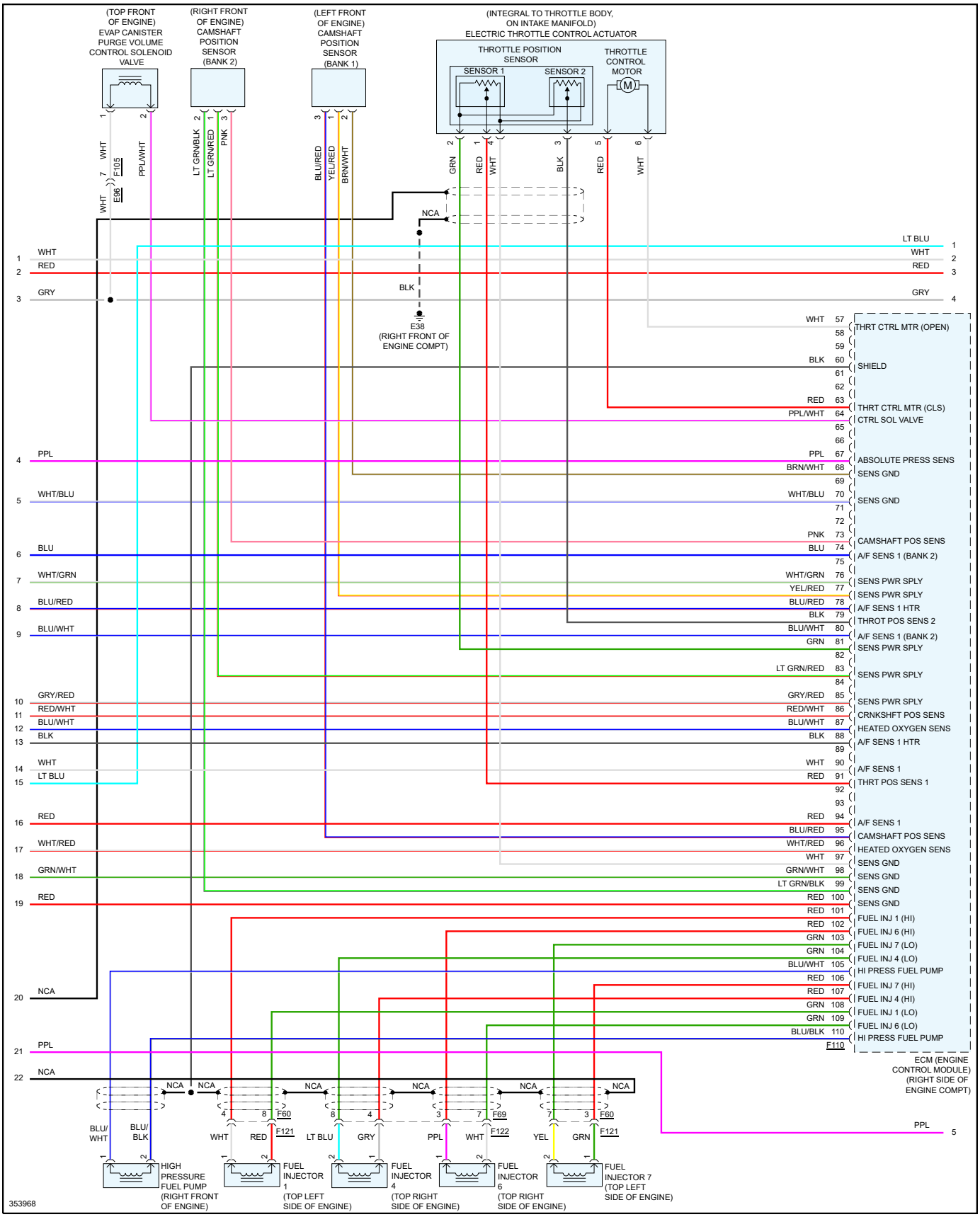
Engine Controls

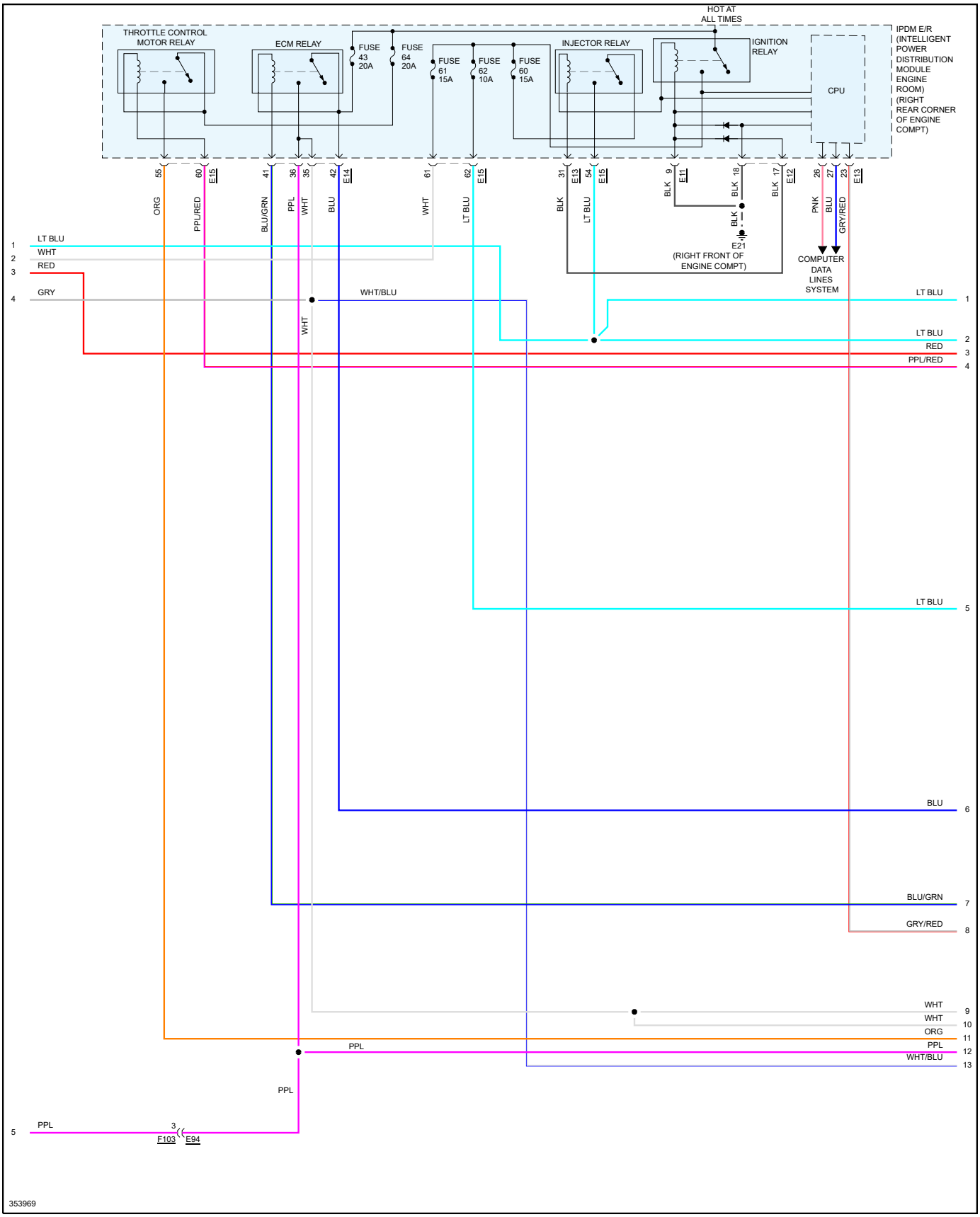
Page 1 of 7

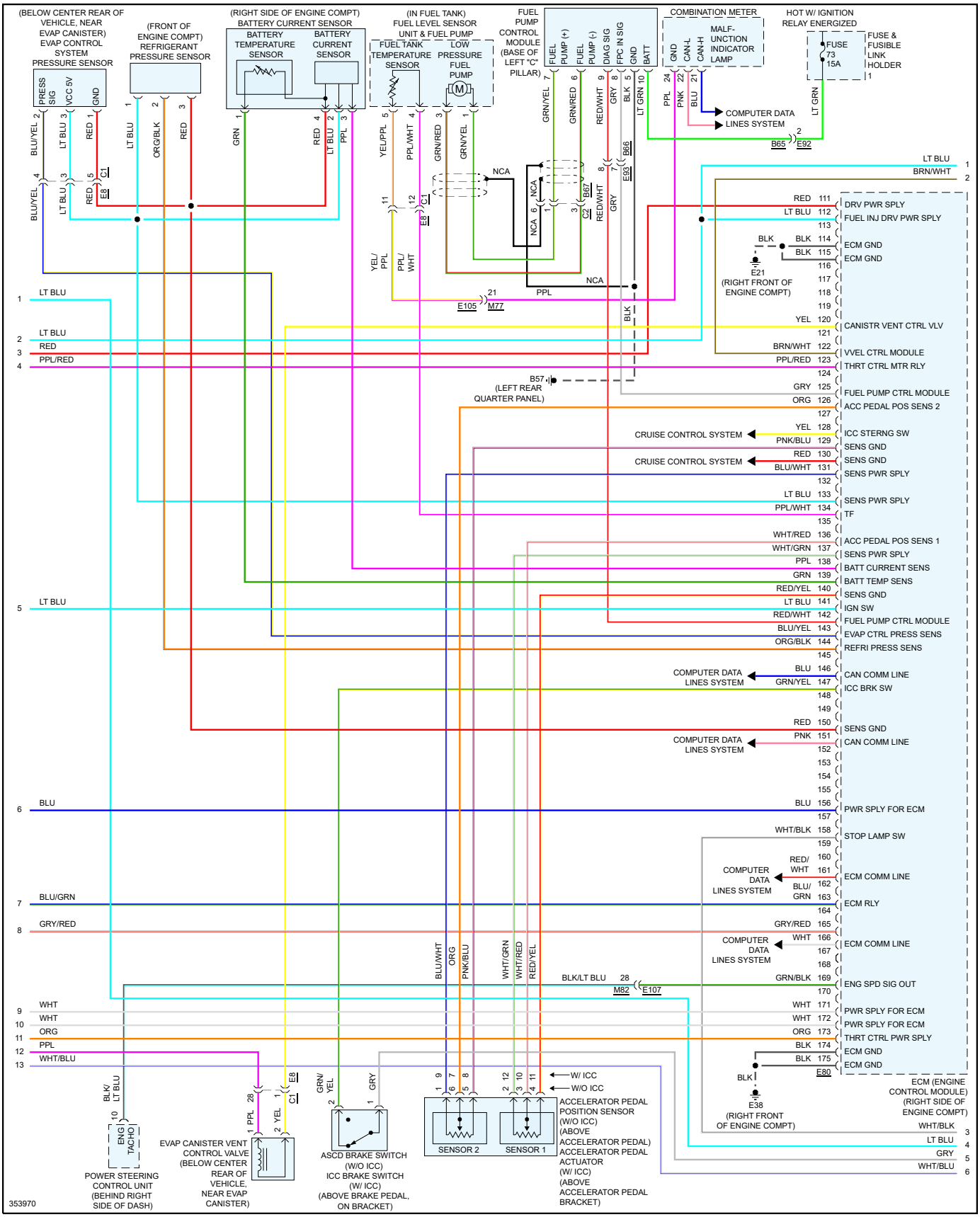

FUEL INJECTOR 8 (TOP RIGHT SIDE OF ENGINE)
FUEL INJECTOR 3 (TOP LEFT SIDE OF ENGINE)
FUEL INJECTOR 2 (TOP RIGHT SIDE OF ENGINE)
FUEL INJECTOR 5 (TOP LEFT SIDE OF ENGINE)
INTAKE TEMPERATURE SENSOR
MASS AIR FLOW SENSOR (LEFT SIDE OF ENGINE COMPT, ATTACHED TO AIR BOX)
(RIGHT FRONT OF ENGINE COMPT) E21
BLK/RED
NCA
F103
E94
E38 (RIGHT FRONT OF ENGINE COMPT)
F105
E96
COOLING FANS SYSTEM
SHIELD
F49
F202
F111
ENGINE CONTROL MODULE (ECM) (RIGHT SIDE OF ENGINE COMPT)
ELECTRONICALLY CONTROLLED COOLING FAN COUPLING
LOW FUEL PRESSURE SENSOR (TOP LEFT OF ENGINE)
POWER STEERING PRESSURE SENSOR (ON POWER STEERING PUMP)
KNOCK SENSOR (BANK 2) (TOP OF ENGINE)
KNOCK SENSOR (BANK 1) (TOP OF ENGINE)
ENGINE OIL TEMPERATURE SENSOR (LOWER LEFT SIDE OF ENGINE)
ENGINE COOLANT TEMPERATURE SENSOR (RIGHT REAR OF ENGINE)
FUEL RAIL PRESSURE SENSOR (LEFT REAR OF ENGINE)
353965

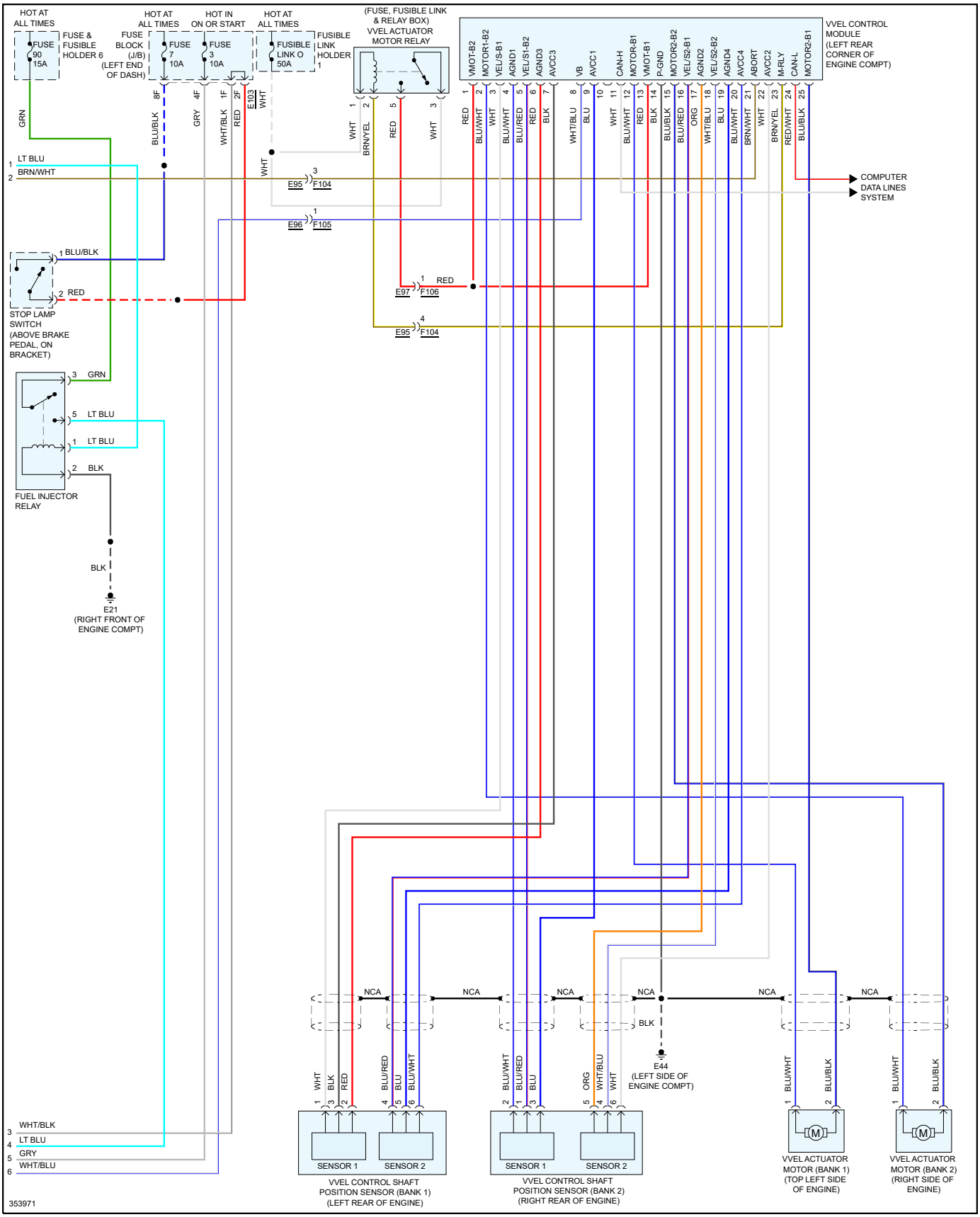

CONDENSER
HOT IN ON OR START
FUSE 71 30A
FUSE & FUSIBLE LINK HOLDER 1
(TOP LEFT SIDE OF ENGINE) IGNITION COIL 1 (W/ POWER TRANSISTOR)
(TOP RIGHT SIDE OF ENGINE) IGNITION COIL 2 (W/ POWER TRANSISTOR)
(TOP LEFT SIDE OF ENGINE) IGNITION COIL 3 (W/ POWER TRANSISTOR)
(TOP RIGHT SIDE OF ENGINE) IGNITION COIL 4 (W/ POWER TRANSISTOR)
WHT
BLK
RED
SPARK PLUG
NCA
BLK
YEL
WHT
BLU/BLK
BRN/YEL
LT GRN/RED
F22
E46
GRY
BRN
YEL
BLU/BLK
BRN/YEL
LT GRN/RED
PNK/BLU
YEL/RED
GRN/YEL
ORG
LT BLU
GRN
BLU/WHT
LT GRN/BLK
BLU/RED
GRN/RED
PPL
COMPUTER DATA LINES SYSTEM
BLU
PNK
BRN
JOINT CONNECTOR
TCM (TRANSMISSION CONTROL MODULE)
A/T ASSEMBLY
INTAKE VALVE TIMING CONTROL SOLENOID VALVE (BANK 1) (LEFT FRONT OF ENGINE)
INTAKE VALVE TIMING CONTROL SOLENOID VALVE (BANK 2) (RIGHT FRONT OF ENGINE)
353966


(TOP LEFT SIDE OF ENGINE) IGNITION COIL 5 (W/ POWER TRANSISTOR)
(TOP RIGHT SIDE OF ENGINE) IGNITION COIL 6 (W/ POWER TRANSISTOR)
(TOP LEFT SIDE OF ENGINE) IGNITION COIL 7 (W/ POWER TRANSISTOR)
(TOP RIGHT SIDE OF ENGINE) IGNITION COIL 8 (W/ POWER TRANSISTOR)
(LOWER REAR OF ENGINE) CRANKSHAFT POSITION SENSOR
(TOP REAR OF ENGINE) MANIFOLD ABSOLUTE PRESSURE SENSOR
NCA
SPARK PLUG
BLK
PNK/BLU
WHT
YEL/RED
GRN/YEL
ORG
GRN/WHT
RED/WHT
WHT/GRN
GRY/RED
WHT/BLU
PPL
BLK
F33
(LEFT SIDE OF ENGINE)
WHT
RED
GRY
BLU
WHT/GRN
BLU/RED
BLU/WHT
RED/WHT
LT BLU
GRN
RED
WHT/RED
GRN/WHT
HEATED OXYGEN SENSOR 2 (BANK 2) (ON OUTLET OF FRONT THREE WAY CATALYST)
HEATED OXYGEN SENSOR 2 (BANK 1) (ON OUTLET OF REAR THREE WAY CATALYST)
AIR FUEL RATIO (A/F) SENSOR 1 (BANK 2) (RIGHT REAR OF ENGINE)
AIR FUEL RATIO (A/F) SENSOR 1 (BANK 1) (LEFT REAR OF ENGINE)
353967


(TOP FRONT OF ENGINE) EVAP CANISTER PURGE VOLUME CONTROL SOLENOID VALVE
(RIGHT FRONT OF ENGINE) CAMSHAFT POSITION SENSOR (BANK 2)
(LEFT FRONT OF ENGINE) CAMSHAFT POSITION SENSOR (BANK 1)
(INTEGRAL TO THROTTLE BODY, ON INTAKE MANIFOLD) ELECTRIC THROTTLE CONTROL ACTUATOR
THROTTLE POSITION SENSOR
SENSOR 1
SENSOR 2
THROTTLE CONTROL MOTOR
E38 (RIGHT FRONT OF ENGINE COMPT)
NCA
THRT CTRL MTR (OPEN)
SHIELD
THRT CTRL MTR (CLS)
CTRL SOL VALVE
ABSOLUTE PRESS SENS
SENS GND
CAMSHAFT POS SENS
A/F SENS 1 (BANK 2)
SENS PWR SPLY
A/F SENS 1 HTR
THROT POS SENS 2
A/F SENS 1 (BANK 2)
CRNKSHFT POS SENS
HEATED OXYGEN SENS
A/F SENS 1
THRT POS SENS 1
CAMSHAFT POS SENS
HEATED OXYGEN SENS
FUEL INJ 1 (HI)
FUEL INJ 6 (HI)
FUEL INJ 7 (LO)
FUEL INJ 4 (LO)
HI PRESS FUEL PUMP
FUEL INJ 7 (HI)
FUEL INJ 4 (HI)
FUEL INJ 1 (LO)
FUEL INJ 6 (LO)
HI PRESS FUEL PUMP
ECM (ENGINE CONTROL MODULE) (RIGHT SIDE OF ENGINE COMPT)
HIGH PRESSURE FUEL PUMP (RIGHT FRONT OF ENGINE)
FUEL INJECTOR 1 (TOP LEFT SIDE OF ENGINE)
FUEL INJECTOR 4 (TOP RIGHT SIDE OF ENGINE)
FUEL INJECTOR 6 (TOP RIGHT SIDE OF ENGINE)
FUEL INJECTOR 7 (TOP LEFT SIDE OF ENGINE)
353968


HOT AT ALL TIMES
THROTTLE CONTROL MOTOR RELAY
ECM RELAY
FUSE 43 20A
FUSE 64 20A
FUSE 61 15A
FUSE 62 10A
FUSE 60 15A
INJECTOR RELAY
IGNITION RELAY
CPU
IPDM E/R (INTELLIGENT POWER DISTRIBUTION MODULE ENGINE ROOM) (RIGHT REAR CORNER OF ENGINE COMPT)
E15
E14
E13
E11
E12
E21
(RIGHT FRONT OF ENGINE COMPT)
COMPUTER DATA LINES SYSTEM
LT BLU
WHT
RED
GRY
WHT/BLU
PPL/RED
BLU
BLU/GRN
GRY/RED
ORG
PPL
F103
E94
353969


(BELOW CENTER REAR OF VEHICLE, NEAR EVAP CANISTER) EVAP CONTROL SYSTEM PRESSURE SENSOR
PRESS SIG
VCC 5V
GND
(FRONT OF ENGINE COMPT) REFRIGERANT PRESSURE SENSOR
(RIGHT SIDE OF ENGINE COMPT) BATTERY CURRENT SENSOR
BATTERY TEMPERATURE SENSOR
BATTERY CURRENT SENSOR
(IN FUEL TANK) FUEL LEVEL SENSOR UNIT & FUEL PUMP
FUEL TANK TEMPERATURE SENSOR
LOW PRESSURE FUEL PUMP
FUEL PUMP CONTROL MODULE (BASE OF LEFT "C" PILLAR)
FUEL PUMP (+)
FUEL PUMP (-)
DIAG SIG
FPC IN SIG
GND
BATT
COMBINATION METER
MALF-UNCTION INDICATOR LAMP
CAN-L
CAN-H
HOT W/ IGNITION RELAY ENERGIZED
FUSE 73 15A
FUSE & FUSIBLE LINK HOLDER 1
COMPUTER DATA LINES SYSTEM
B65
E92
NCA
B57 (LEFT REAR QUARTER PANEL)
E105
M77
E21 (RIGHT FRONT OF ENGINE COMPT)
DRV PWR SPLY
FUEL INJ DRV PWR SPLY
ECM GND
CANISTR VENT CTRL VLV
VVEL CTRL MODULE
THRT CTRL MTR RLY
FUEL PUMP CTRL MODULE
ACC PEDAL POS SENS 2
ICC STERNG SW
SENS GND
SENS PWR SPLY
TF
ACC PEDAL POS SENS 1
BATT CURRENT SENS
BATT TEMP SENS
IGN SW
EVAP CTRL PRESS SENS
REFRI PRESS SENS
CAN COMM LINE
ICC BRK SW
PWR SPLY FOR ECM
STOP LAMP SW
ECM COMM LINE
ECM RLY
ENG SPD SIG OUT
THRT CTRL PWR SPLY
E80
E38 (RIGHT FRONT OF ENGINE COMPT)
ECM (ENGINE CONTROL MODULE) (RIGHT SIDE OF ENGINE COMPT)
M82
E107
POWER STEERING CONTROL UNIT (BEHIND RIGHT SIDE OF DASH)
ENG TACHO
EVAP CANISTER VENT CONTROL VALVE (BELOW CENTER REAR OF VEHICLE, NEAR EVAP CANISTER)
ASCD BRAKE SWITCH (W/O ICC) ICC BRAKE SWITCH (W/ ICC) (ABOVE BRAKE PEDAL, ON BRACKET)
ACCELERATOR PEDAL POSITION SENSOR (W/O ICC) (ABOVE ACCELERATOR PEDAL) ACCELERATOR PEDAL ACTUATOR (W/ ICC) (ABOVE ACCELERATOR PEDAL BRACKET)
SENSOR 2
SENSOR 1
W/ ICC
W/O ICC
353970


HOT AT ALL TIMES
FUSE & FUSIBLE HOLDER 6
FUSE 90 15A
HOT AT ALL TIMES
HOT IN ON OR START
HOT AT ALL TIMES
FUSE BLOCK (J/B) (LEFT END OF DASH)
FUSE 7 10A
FUSE 3 10A
FUSIBLE LINK O 50A
FUSIBLE LINK HOLDER 1
(FUSE, FUSIBLE LINK & RELAY BOX)
VVEL ACTUATOR MOTOR RELAY
VVEL CONTROL MODULE (LEFT REAR CORNER OF ENGINE COMPT)
VMOT-B2
MOTOR1-B2
VEL/S-B1
AGND1
VEL/S1-B2
AGND3
AVCC3
VB
AVCC1
CAN-H
MOTOR-B1
VMOT-B1
P-GND
MOTOR2-B2
VEL/S2-B1
AGND2
VEL/S2-B2
AGND4
AVCC4
ABORT
AVCC2
M-RLY
CAN-L
MOTOR2-B1
COMPUTER DATA LINES SYSTEM
STOP LAMP SWITCH (ABOVE BRAKE PEDAL, ON BRACKET)
FUEL INJECTOR RELAY
E21 (RIGHT FRONT OF ENGINE COMPT)
E95
F104
E96
F105
E97
F106
E44 (LEFT SIDE OF ENGINE COMPT)
NCA
SENSOR 1
SENSOR 2
VVEL CONTROL SHAFT POSITION SENSOR (BANK 1) (LEFT REAR OF ENGINE)
VVEL CONTROL SHAFT POSITION SENSOR (BANK 2) (RIGHT REAR OF ENGINE)
VVEL ACTUATOR MOTOR (BANK 1) (TOP LEFT SIDE OF ENGINE)
VVEL ACTUATOR MOTOR (BANK 2) (RIGHT SIDE OF ENGINE)
353971